

Systematic Data Cleaning & Audit Log (Cleaning Log)

Course: Quantitative Data Analysis (M1 – S7) **Coursework Deliverable:** Artefact 1 – Audit Component (Slide 39 & 41) **Dataset:** S&P 500 Corporate Governance & Financial Performance (FY2025) **Total Records Audited:** $N = 503$ observations, 17 raw features

1. Executive Cleaning Summary

In professional consulting and econometric modeling, an unexamined dataset produces indefensible conclusions. As established in the course doctrine (*“Garbage in, garbage out. Software never refuses to print a number. Sanity-checking output is your job”* – Slides 25 & 34), this log systematically documents all anomalies identified, the corrective actions taken, and the methodological justifications.

Total Raw Rows Extracted:	503
Total Cleaned Analytical Rows:	503
Duplicate Tickers / Rows:	0 (100% Unique)
Remediated Ticker Entities:	1 (Fiserv, Inc.)
Categorical Formatting Fixes:	503 (Standardized to 5 GICS groups)
Decimal / Type Harmonization:	100% Validated (Period decimal separators, numeric floats)

2. Granular Audit Matrix: What Was Wrong, What Was Done, Why

#Inspection Category (Slide 39)	Anomaly Detected in Raw Data	Corrective Action Taken	Methodological & Business Justification
1 Missing Values in Disguise	Row (FISV) returned NaN across sector, turnover, margin, and governance scores.	Identified corporate ticker transition from FISV to FI on NYSE. Programmatically re-mapped to FI and populated audited FY2025 financials and ISS risk scores.	Prevents silent data attrition. Fiserv is an S&P 500 fintech market leader (\$19.09B revenue); retaining it eliminates sample truncation bias.
2 Duplicate Rows	Evaluated ticker column uniqueness (df['Ticker'].is_unique).	Zero duplicate rows detected (503 unique equity symbols).	Guarantees that each observation represents exactly one independent firm.
3 Imposable/Extreme Values: ROE	Masco Corp exhibited an ROE of 58.625 (5,862.5%).	Retained raw observation without arbitrary deletion or trimming. Documented in statistical tables that Mean = 39.5% is distorted and declared the Median (16.8%) as the official reporting metric.	Preserves empirical authenticity. Trimming eliminates real accounting phenomena (Masco had \$80M equity vs \$1B net income). Using median protects against outlier distortion (Slide 21).
4 Structural Missingness: Negative Equity ROE	32 blue-chip corporations (e.g., AbbVie, Altria, AutoZone, Booking, Domino's, HP Inc.) had NaN for ROE.	Evaluated balance sheets; confirmed negative book equity driven by cumulative share repurchases (<i>treasury stock buybacks</i>). Retained ROE for $N = 471$ positive-equity firms, and established Net Profit Margin ($N = 503$ valid) as the primary cross-firm profitability benchmark.	Deleting 32 premier firms would cause severe survivorship and industry bias (Retail, Tobacco, Pharma). Profit Margin is unaffected by capital structure and valid across all 503 firms.
5 Inconsistent Categories	Raw data contained 11 distinct GICS sectors, exceeding the mandatory course limit of 2 to 5 nominal categories.	Mapped all 11 sectors into 5 homogeneous business groups (Tech & Comms, Healthcare, Finance, Industrials & Energy, Consumer).	Mandatory requirement for ANOVA (Session 3) and Chi-Square (Session 2). Prevents statistical fragmentation and ensures adequate cell frequencies ($n \geq 60$ per sector).
6 Variability: Text & Figures for Annotations	Risk column of nominal categories specified a household characteristic for annotations.	Standardized all financial metrics, removed whitespace, and formatted variables: Market_Cap, ROE, Profit_Margin, Governance_Risk_Level.	Prevents misclassification of variables as accents. Standardized formatting ensures consistent interpretation across slides.

3. Data Completeness Profile (Before vs. After Cleaning)

Variable Name	Jamovi Type	Valid Observations (Raw)	Valid Observations (Cleaned)	Completeness Rate	Treatment Rationale
Ticker	ID	503	503	100.0%	Primary key, unique
Company	Nominal	503	503	100.0%	Complete
Sector	Nominal (5 groups)	502	503	100.0%	Fiserv mapped to <i>Finance</i>
Total_Revenue_B	Continuous	502	503	100.0%	Fiserv updated (\$19.09B)
Market_Cap_B	Continuous	503	503	100.0%	Complete
Market_Cap_Quartile	Ordinal (4 tiers)	503	503	100.0%	Perfect quartile distribution ($n \approx 126$)
Profit_Margin	Continuous	502	503	100.0%	Primary uncorrupted profitability metric
ROE	Continuous	471	471	93.6%	32 firms negative equity (transparently reported)
Beta	Continuous	498	499	99.2%	Standard 5Y monthly volatility
Headcount	Continuous	499	500	99.4%	Full-time employees
Overall_Governance_Risk	Continuous	495	496	98.6%	ISS composite score (1–10)
Governance_Risk_Level	Ordinal (3 tiers)	495	496	98.6%	Low ($n = 150$), Med ($n = 149$), High ($n = 197$)

4. Verification Check against Costly Student Traps (Slide 42)

- ☒ **Trap 1: Reporting a mean for a nominal variable:** Avoided. Sector is analyzed strictly via frequency distributions and mode.
- ☒ **Trap 2: Setting the measure type wrong in Jamovi:** Avoided. Every variable has a documented Jamovi Measure Type in Codebook_Jamovi.
- ☒ **Trap 3: Reading “Sig. = 0.000” as $p = 0$:** Avoided. Formatted rigorously in APA style as $p < .001$.

- ☒ **Trap 4: Confusing statistical significance with strength:** Addressed. We report effect sizes (*IQR*, *SD*, Skewness, Kurtosis) alongside hypothesis tests.
- ☒ **Trap 5: Copying output with no interpretation:** Addressed. Full narrative interpretations written for all variables.
- ☒ **Trap 6: Stopping at the statistic:** Addressed. Formulated executive Monday-morning recommendations.